

# RESHMA THOMAS

AI / Machine Learning Engineer | Applied GenAI | Document Intelligence  
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## PROFESSIONAL SUMMARY

AI/ML Engineer with end-to-end experience designing, training, and deploying machine learning systems that solve real-world business problems. Proficient in Python, TensorFlow, and PyTorch, with a proven track record building production-grade AI solutions using LLMs, Vision-Language Models, OCR pipelines, and computer vision architectures. Strong ability to translate complex AI capabilities into practical, measurable business outcomes.

## TECHNICAL SKILLS

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|------------------------|----------------------------------------------------------------------------------|
| Languages              | Python (primary), SQL, Bash                                                      |
| ML / DL Frameworks     | TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV                                 |
| LLMs & GenAI           | Gemini 1.5 Pro, OpenAI API, Prompt Engineering, RAG, Fine-tuning                 |
| Vision-Language Models | Qwen 2.5 VLM — benchmarking, inference, document reasoning                       |
| Computer Vision        | CNNs (MobileNet), YOLO, COCO, Object Detection & Segmentation                    |
| NLP                    | NLTK, spaCy, Transformers, Text Classification, Named Entity Recognition         |
| OCR & Doc Intelligence | PaddleOCR, EasyOCR, Structured Data Extraction, Multi-language Pipelines         |
| Data Engineering       | Pandas, NumPy, ETL Pipelines, Feature Engineering, Dataset Annotation (Roboflow) |
| Data Visualisation     | Matplotlib, Seaborn, Analytics Dashboards                                        |
| Backend & APIs         | FastAPI, Flask, Django, RESTful API Design, Telegram Bot API                     |
| Deployment             | Docker, Streamlit, Google Colab, Inference Optimisation, Latency Tuning          |
| Dev Tools              | Git/GitHub, VS Code, PyCharm, Jupyter Notebook, Anaconda, Roboflow               |

## AI / ML PROJECTS

### Medicine Blister Text Extraction — PaddleOCR, EasyOCR, Qwen 2.5 VLM, Roboflow

- Built end-to-end document intelligence pipeline to extract and validate text from medicine blister packaging with automated expiry date detection.
- Benchmarked open-source VLMs against traditional OCR; findings influenced the project's architectural roadmap.
- Performed full dataset curation, annotation, and preprocessing pipeline design using Roboflow.

### AI Credit Card Expense Analyzer — Gemini 1.5 Pro, FastAPI, Docker

- Designed full-stack AI system to parse unstructured financial PDFs and extract structured transaction data, eliminating manual data entry.
- Built FastAPI REST backend; containerised with Docker for scalable, production-ready deployment.

### **Voice for the Voiceless — Multimodal AI** — *MobileNet CNN, OpenCV, Streamlit, Telegram Bot API*

- Fine-tuned CNN facial expression recognition model to 91% accuracy; validated 89% real-time inference accuracy via live webcam testing.
- Deployed Streamlit application integrating real-time CV inference with automated Telegram caregiver alerts.

### **Find My Stress — AI Web Application** — *SVM, Django, Python*

- Built full-stack web application integrating SVM predictive model (91.05% accuracy) with an AI chatbot for personalised student mental health support.

### **Object Segmentation for Semi-Humanoid Robotics** — *YOLO, COCO, Roboflow*

- Developed real-time object detection and segmentation pipelines for industrial logistics automation.
- Curated and annotated custom datasets; benchmarked model variants to optimise inference latency for production deployment.

### **STAR KIDS — Full-Stack Web Application** — *Flask, SQLite, Analytics Dashboard*

- Built full-stack gamified web application with real-time data analytics dashboard — demonstrating end-to-end product delivery capability.

## **PROFESSIONAL EXPERIENCE**

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### **AI / ML Engineer Intern** | **i Hub Research & Robotics Pvt. Ltd** | *Oct 2024 – Present*

- Led development of medicine blister OCR and VLM pipeline; independently researched open-source frontier models and influenced project architectural decisions.
- Developed YOLO-based computer vision pipelines for semi-humanoid robotics; curated datasets, trained models, and benchmarked variants for production deployment.
- Collaborated across engineering and research teams to optimise inference accuracy and latency.

### **Contributing Editor & Editorial Council Member** | **Shalom Tidings, Shalom Media USA** | *Jan 2017 – Present*

- Designed and deployed automated content workflows processing 1,000+ documents annually across 12 languages — delivering 40% reduction in processing time.
- Built advanced tracking and analytics systems; managed coordination across 50+ global contributors.
- Gold Award, Australian Religious Press Association (2020); Best Magazine of the Year — Catholic Press Awards USA (2021, 2022, 2023).

## **EDUCATION**

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**Post Graduate Diploma in Artificial Intelligence & Machine Learning** — I Hub School of Learning (2025)

**B.Tech in Biotechnology, First Class with Honors** — Calicut University, Kerala

**Diploma in Administration and Management** — ICFAI University, Dehradun

## **CERTIFICATIONS**

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- Machine Learning Specialization — Stanford University / DeepLearning.ai (Coursera)
- Python 101 for Data Science — IBM Cognitive Class
- Introduction to Programming and Python — Kaggle

## **ADDITIONAL INFORMATION**

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**Languages:** English (fluent), Malayalam (native)

**Interests:** Applied AI for social impact, healthcare AI, enterprise automation, and generative AI research.